Vihal Vemula

Lead Android Architect

Phone: (336) 441-5664 **Email:** svihal07@gmail.com

Profile Summary:

- **With over a decade of experience in Android application development and 12+ years of IT exposure, I specialize in delivering versatile, high-quality solutions for diverse devices, including smartphones, tablets, Android smartwatches (Wear OS), and smart TVs.** Proficient in Kotlin and Java, I have expertise in utilizing advanced methodologies such as dependency injection, test-driven development (TDD), and Jetpack to craft scalable applications.
- I bring extensive knowledge of RESTful web services, coupled with expertise in tools like Retrofit, Volley, and OKHTTP. Additionally, I am proficient in web technologies like JavaScript, HTML, JSON, AJAX, XML, and SOAP, ensuring smooth integration with web-based services.
- With a focus on quality assurance, I excel in automated testing frameworks such as Espresso, Robolectric, and Jenkins. My experience spans Android BLE, Google Fit, and Android TV development, with deep proficiency in Jetpack Compose, Android design patterns, and scalable architectures. I also prioritize accessibility by implementing ADA-compliant features to create inclusive user experiences.
- My background also includes working with Flutter and Dart, alongside a strong foundation in mobile development. I integrate IoT technologies into my solutions, leveraging interconnected systems to build innovative products and ensuring seamless transitions from Java to Kotlin for improved code quality and performance.
- I thrive in Agile environments, actively contributing to sprint planning, daily stand-ups, and backlog management. Skilled in Git, BitBucket, and SourceTree, I follow best practices for source control. My work aligns with Google Material Design principles to create intuitive user interfaces.
- Beyond Android, I possess hands-on expertise in IoT and M2M applications, using technologies like GSM/GPRS modules, Bluetooth, and cellular connectivity for innovative solutions in automation and security. With experience in IntelliJ, Eclipse, and Android Studio, I continually embrace emerging technologies to deliver impactful results.

Technical Skills:

Android Platforms: Android Studio, Firebase, Android JetPack, Jetpack Compose, Android SDK, Content Resolvers, RecyclerView Layout Manager, RecyclerView Animator, Location Services, Content Providers, JobScheduler, Work Manager, Espresso, Broadcast Receiver, Services, Activities, LiveData, Kotlin Flows, ViewModel, Room DB, Datastore
Programming Languages: Kotlin, Java, Python, Groovy, Kotlin DSL, XML, JavaScript
Web Development: HTML, CSS, JavaScript, React.JS
Frameworks: Firebase Push Notifications, Firebase Firestore, Google Places, Google Play Services, Google Maps API, Volley, Retrofit, JUnit, Kotlin Coroutines, Kotlin Flow, React.JS
Architectures and Design Patterns: MVC, MVP, MVVM, CLEAN, MVI, Builder, Factory, Singleton, Adapter, Observer
Android Development: Constraints, Handlers, Threads, Loopers, Loaders, Java, RxJava, Picasso, Glide, Coil, Custom Android Views, ListView, RecyclerView, Widgets (Jetpack Glance), Fragments, Bundle, Intents, Runtime Permissions, PendingIntent, Repository Layer, Single Source of Truth, Cache, Offline Mode
Servers & Databases: SQL, Firebase Firestore, SQLite, GraphQL, MongoDB
IDEs: Android Studio, IntelliJ, SourceTree, GitHub Desktop

Issue Tracking & Project Management Tools: JIRA, Jenkins, JUnit, Unit Testing, Agile, Scrum, Git, SVN, GitHub, BitBucket, GitLab, Travis, Confluence, GitHub Actions
Third-Party Frameworks/APIs: Glide, Picasso, Coil, Google Console APIs, Samsung SDK, Logger, Dagger, Moshi, Robolectric, JUnit, LeakCanary, Robotium, NFC (Near Field Communication) technology, Retrofit, OkHttp, Http Logging Interceptor, Certificate Pinning
Tools: DDMS, ADB, LeakCanary, Firebase, Firebase Crashlytics, ART, Bluetooth Low Energy, Test Driven Development, Continuous Integration, Android Profiler
UI Components: Material Design

Professional Experience:

September 2023 - Present **Lead Android App Developer**
Truist Financial Corporation, Charlotte, North Carolina

https://play.google.com/store/apps/details?id=com.truist.mobile&hl=en-US

Summary: As a Lead Android Developer at Truist Financial, I led the modernization and security enhancement of the mobile banking app, integrating biometric authentication and multi-factor verification for robust user protection. I optimized app performance using Jetpack Compose, Kotlin Coroutines, and Flow while ensuring compliance with financial security standards. My contributions included implementing advanced encryption protocols, fortifying data security, and delivering a seamless, intuitive user experience through collaboration with cross-functional teams.

- Integrate advanced biometric authentication features, including fingerprint and facial recognition, to ensure secure and seamless access to Truist Financial's app, prioritizing user safety and data integrity.
- Implement clean code principles, adhering to Uncle Bob's guidelines, and utilize the MVVM architectural pattern within a TDD setup to enhance the scalability and maintainability of the app's network and presentation layers.
- Transition legacy projects to leverage Jetpack Compose, modernizing the app's UI/UX and improving both developer efficiency and user satisfaction.
- Utilize Kotlin Flow and Coroutines for reactive programming, optimizing asynchronous operations to improve responsiveness and streamline critical app functionalities.
- Collaborate on the development and rollout of multi-factor authentication (MFA) mechanisms, adding robust layers of security to user logins and reducing vulnerabilities.
- Design and implement encryption strategies to secure sensitive user data, ensuring compliance with financial data regulations and minimizing risks associated with breaches or unauthorized access.
- Enhance app functionality and performance using libraries and tools such as Firebase, Google Maps SDK, ReactiveX, and Picasso, while ensuring seamless user experiences.
- Strengthen app security by integrating tools like RootBeer for root detection and Mitek MiSnap for secure document capture, mitigating potential security threats.
- Partner with backend teams to implement secure server-side communication protocols, safeguarding data integrity across all client-server interactions and meeting stringent financial standards.
- Conduct unit, integration, and penetration tests in collaboration with QA teams to identify and resolve potential vulnerabilities, ensuring a robust and secure mobile app.
- Manage Jetpack Compose lifecycle events using Compose's own Runtime layer with tools like remember and LaunchedEffect, effectively handling side effects and resource management for a smooth user experience.
- Implement real-time analytics using Airship Performance Analytics, Segment, and Azure Event Hubs, enabling data-driven decisions and performance optimization for the mobile app.
- Actively participate in Agile practices, including sprint planning and daily stand-ups, ensuring efficient feature delivery in a DevOps-driven development environment.
- Work with cross-disciplinary teams to integrate permission-controlled foreground services, supporting uninterrupted background processes essential to user workflows.
- Align app development with critical accessibility guidelines to enhance usability for all users.

- Integrate advanced real-time fraud detection algorithms to protect against unauthorized transactions.
- Implement secure key storage using the Android Keystore System to safeguard sensitive information.
- Implement secure token-based authentication for API interaction, ensuring robust communication protocols.

Oct 2021 – Aug 2023 **Lead Android Architect**
New Balance, Boston, Massachusetts

https://play.google.com/store/apps/details?id=pl.newbalance.club&hl=en-US
Summary: As the Lead Android Architect at New Balance, I led the development and optimization of the *New Balance Club* loyalty app, enhancing user engagement through features like exclusive deals, pre-order access, and personalized notifications. I drove the transition to modern technologies such as Jetpack Compose, improving performance, security, and cross-platform consistency. Additionally, I integrated advanced tools like Okta, Firebase, and Stripe, ensuring seamless functionality and a top-tier user experience.

- Led the development and upkeep of the New Balance Club loyalty app, ensuring a seamless, intuitive user experience by leading the creation of features such as pre-order access, exclusive deals, and personalized notifications.
- Optimized the integration of essential features like access to limited products, free shipping, extended return windows, and exclusive offers, while ensuring the app maintains performance standards.
- Directed the creation of tailored offers and notifications using user data, ensuring customers receive relevant information regarding New Balance specials, new product launches, and sales events.
- Developed cross-platform mobile applications using React Native, ensuring seamless integration with Android-specific functionalities such as native modules, custom views, and push notifications.
- Enhanced security by configuring Okta's multi-factor authentication, integrating it seamlessly with the app's login system.
- Leveraged Kotlin Coroutines and Flow API alongside Retrofit, OkHttp, NavGraph, deep linking, and paging to manage data flows efficiently.
- Implemented secure login protocols by utilizing the Biometric API for enhanced authentication processes.
- Collaborated with UI/UX designers to bring Figma mockups to life, transforming them into dynamic user interfaces using Jetpack Compose.
- Adopted Jetpack Compose for server-driven UIs, transitioning from traditional XML layouts to a modern Compose approach to improve design flexibility and maintainability.
- Integrated Hilt for dependency injection, optimizing the app's architecture and streamlining code maintenance and scalability.
- Led the shift from RESTful API to GraphQL with Apollo Client, resulting in optimized data fetching and improved app performance.
- Optimized application performance on Android by leveraging platform-specific code with React Native's Native Modules and bridging functionalities for complex use cases.
- Implemented the MVVM (Model-View-ViewModel) architecture pattern to create modular, testable, and maintainable Android applications, utilizing LiveData and ViewModel for managing UI-related data lifecycle-aware components and ensuring a clear separation of concerns.
- Utilized Firebase Realtime Database for real-time data synchronization and Firebase Remote Config to dynamically adjust the app's settings and content.
- Ensured data security through real-time updates in Google Firestore, incorporating robust authentication mechanisms and user ID validation.
- Collaborated with UI/UX teams to seamlessly integrate personalized recommendation features within the app, enhancing user experience and engagement.
- Implemented location-based services using the Google Maps SDK and added engaging animations using Lottie for a richer app experience.
- Integrated Stripe Payment SDK to facilitate secure and efficient payment processing within the app's UI.
- Managed periodic data synchronization with WorkManager, ensuring reliable background tasks for data syncs, analytics reporting, and log transmission to backend services.

- Improved build efficiency by fine-tuning Gradle configurations within Azure, reducing build times and optimizing the deployment pipeline.
- Employed Clean Code principles and the MVI (Model-View-Intent) Reactive Architecture, establishing a structured, unidirectional data flow to improve scalability and maintainability.
- Modularized the customer payment authentication flow by implementing pair programming techniques and integrating third-party SDKs to enhance payment security.
- Implemented Splunk MINT SDK to collect critical crash, performance, and usage data, enhancing app analytics and debugging capabilities.
- Utilized Git for version control and streamlined CI/CD processes with Azure DevOps, enhancing collaboration and ensuring smooth code reviews and deployments.
- Conducted comprehensive testing of Jetpack Compose components, using the Semantic API to ensure stability through unit and integration tests.

Mar 2020 – Sep 2021 **Sr. Android Developer**
Sun Country Airlines, Minneapolis, Minnesota

https://play.google.com/store/apps/details?id=com.suncountry.suncountrymobile&hl=en_IN
Summary: As a Sr. Android Developer with Sun Country Airlines, I spearheaded the design and development of cutting-edge mobile solutions aimed at improving the passenger experience. With expertise in Kotlin and Flutter, I crafted seamless, high-performance features such as payment systems, real-time flight tracking, and trip management tools. My focus has been on optimizing performance, upholding stringent security protocols, and delivering a smooth, intuitive experience for travellers.

- Developed and enhanced key features for Sun Country Airlines, such as the SkyAir Fast Checkout for streamlined payments with stored credit card details and Dynamic Boarding Queue for real-time seat readiness notifications.
- Created essential travel tools, including real-time flight tracking, boarding passes, and check-in reminders, all displayed on the home screen and powered by Kotlin Flows for real-time updates.
- Improved trip management functionalities—seat selection, itinerary changes, and additional service options—by utilizing Kotlin Coroutines for efficient task handling.
- Built custom packages in Flutter, leveraging native Android and iOS APIs to elevate user experience.
- Partnered with UI/UX designers to design intuitive, visually appealing interfaces for time-sensitive features like boarding passes.
- Ensured consistency with brand guidelines and maintained cohesive updates across interfaces.
- Implemented offline functionality using Room Database, allowing access to critical boarding information without internet connectivity.
- Integrated geolocation features through Flutter Geolocator for precise location tracking.
- Applied top-tier security measures to safeguard user data and ensure compliance with industry standards.
- Advocated for Test-Driven Development (TDD) with JUnit and Espresso to maintain high-quality, bug-free code.
- Utilized Firebase Crashlytics for real-time monitoring and resolution of production crashes, improving app stability.
- Enhanced backend communication with Kotlin Coroutines, Retrofit, and OkHttp for optimized data processing.
- Applied Agile methodologies—SCRUM, sprint planning, retrospectives, and standups—to ensure on-time delivery of features.
- Collaborated with product managers, designers, and backend teams via JIRA, Slack, and Teams to ensure seamless coordination.
- Managed version control and CI/CD pipelines using GitHub and GitLab, maintaining a clean and efficient codebase.
- Implemented MVVM architecture to ensure a scalable codebase, utilizing Kotlin Coroutines for efficient background task management and data handling.
- Integrated Omniture Tracking for real-time data collection and actionable business insights.
- Provided technical support to quality analysts to ensure thorough test coverage.

- Stayed up to date with the latest Kotlin and Android best practices to maintain a modern approach.
- Mentored junior developers, conducted code reviews, and guided the team in following best practices with Kotlin and Coroutines.
- Refined key modules into a feature-based multi-module structure to accelerate builds and simplify testing.
- Initiated a Flutter parallel project for rapid testing of new UI concepts, enabling faster prototyping and experimentation.

Dec 2018 – Feb 2020 **Sr. Android Engineer**
Honda Motor Company, Troy, OH

https://play.google.com/store/apps/details?id=com.honda.accessories.genuine&hl=en-US
Summary: As a Sr. Android Engineer at Honda, I led the design and enhancement of the Honda app, ensuring seamless integration with Honda's connected vehicles. I developed key features such as remote vehicle access, temperature control, and intelligent trip planning with charging station recommendations. Additionally, I applied Android Jetpack components and MVVM architecture to ensure scalability and performance, while mentoring junior developers and collaborating with cross-functional teams to deliver high-quality app solutions.

- Directed the design and improvement of the Honda mobile app, ensuring smooth and reliable integration with Honda's connected vehicle models.
- Enabled mobile phone functionality as a secure key for locking/unlocking vehicles remotely and managing vehicle access.
- Implemented remote vehicle temperature control, allowing users to warm up or cool down their vehicle before entering.
- Developed intelligent trip planning features, offering recommended routes and charging stops tailored to user preferences.
- Integrated map functionalities for users to discover nearby charging stations and points of interest, with the ability to send destinations directly to their vehicle.
- Led the transition of web-based components to native Android solutions using technologies like Kotlin, Android Studio, and native SDKs.
- Architected maintainable mobile applications leveraging Android Jetpack components and the MVVM pattern for scalable development.
- Applied core Android Architecture Components such as LiveData, ViewModel, and Room to implement the MVVM design pattern for robust app development.
- Enhanced app performance by integrating third-party SDKs, focusing on real-time data tracking and improved device interactions.
- Utilized Dagger 2 for implementing modular architecture, enabling better feature separation and future scalability.
- Managed asynchronous tasks and multithreading with RxKotlin, RxAndroid, and RxBinding to optimize app performance.
- Optimized UI/UX performance by designing lightweight layouts using RecyclerView and ConstraintLayout, reducing nesting and improving responsiveness.
- Handled activity lifecycle management using AndroidX Activity to ensure proper resource allocation and state persistence.
- Implemented multimedia features using the Android Jetpack Media API to enable video and audio recording, playback, and streaming functionalities.
- Ensured compliance with automotive app standards, maintaining high levels of security and user privacy.
- Provided mentorship to junior developers, offering technical leadership and guidance on best development practices.
- Conducted code reviews and debugging sessions, focusing on maintaining high standards of code quality and application stability.
- Wrote unit tests using JUnit and Mockito to verify functionality and prevent regressions in the application.
- Monitored build progress and deployment cycles using Azure DevOps, tracking code quality with integrated dashboards.

- Collaborated with cross-functional teams to align on project goals, meeting deadlines while ensuring high-quality outcomes.

Apr 2017 – Nov 2018 **Android Developer**
IKEA, Conshohocken, Pennsylvania

https://play.google.com/store/apps/details?id=com.ingka.ikea.app&hl=en-US
Summary: As an Android Developer with IKEA, I contributed to the development and optimization of key features in the IKEA mobile app, enhancing the shopping experience both online and in-store. I implemented features such as "Scan & Shop" for seamless in-store purchases, real-time order tracking, and personalized product lists. Additionally, I integrated IKEA Family benefits and ensured secure app functionality while optimizing performance for a smooth user experience.

- Pairprogram with development team and enhancement of key features for the IKEA mobile app, improving product discovery and providing a seamless shopping experience both online and in-store.
- Integrated the "Scan & Shop" functionality, enabling users to scan products while shopping in-store and skip the checkout process.
- Designed and maintained features that allow users to save products in personalized lists, facilitating future purchases and project planning.
- Implemented real-time delivery tracking, offering users the ability to follow their orders and receive updates directly through the app.
- Optimized the integration of IKEA Family perks, providing users with quick access to their membership card and past receipts.
- Migrated legacy Java codebase to Kotlin, improving code readability, reducing boilerplate code with Kotlin's concise syntax, and enhancing application performance and reliability.
- Developed secure token-based authorization to protect access to sensitive features within the app.
- Enabled instant communication through push notifications and Google Cloud Messaging, ensuring timely updates within the IKEA app.
- Applied security measures using Dexguard to safeguard the app from reverse engineering and potential vulnerabilities.
- Utilized local storage solutions like SQLite to manage persistent user data, focusing on preferences and scheduling features.
- Refined the app's codebase to ensure better reusability, optimize performance, and reduce app size.
- Coordinated version control and collaboration through private GitHub repositories, ensuring efficient code integration and stability.
- Contributed to Agile sprint planning and daily standups, ensuring the timely delivery of features and smooth workflow.
- Integrated backend communication through Retrofit and RxJava, enhancing data exchange within the app and improving responsiveness.
- Developed modular business logic using design patterns like Singleton, Builder, and Proxy to enhance app functionality.
- Improved dependency injection and app architecture by transitioning to Dagger for better class instance management.
- Implemented robust unit tests by mocking API responses with tools like Mockito, ensuring app stability and performance.
- Created dynamic navigation elements within the app, integrating activities and fragments for smooth user experiences.
- Enhanced the user interface by incorporating a collapsing toolbar, creating a more intuitive design while users scroll through content.
- Managed issue tracking, bug fixes, and feature requests using GitHub Issues, ensuring efficient resolution and progress in development.

Jan 2015 – Mar 2017 **Android Engineer**
Mint Mobile, Fountain Valley, CA

https://play.google.com/store/apps/details?id=com.uvnv.mintsim&hl=en-US
Summary: As an Android Engineer at Mint Mobile, I led the development and optimization of key features in the Mint Mobile app, including quick refills, auto-pay management, and real-time usage tracking. I enhanced user security with password-free login options like Face Recognition and Fingerprint Authentication, while integrating seamless payment systems and device activation functionalities.

- Developed and optimized key features in the Mint Mobile app, including quick refills, auto-pay management, and real-time usage tracking.
- Implemented secure, password-free login options using Face Recognition and Fingerprint Authentication for improved user experience.
- Enhanced account management features, allowing users to manage notifications, reset passwords, and update email addresses.
- Integrated payment systems for seamless bill payments and account management.
- Enabled device activation via the app, allowing users to activate new numbers, port-in requests, and manage plans.
- Implemented barcode/QR code scanning for easy payment and service activation.
- Developed RESTful APIs for data fetching and integration with GSON.
- Enhanced performance using OEM SDKs and RxJava 2 for optimized app responsiveness.
- Transitioned from Dagger 1 to Dagger 2 for improved dependency injection and app architecture.
- Integrated advanced camera/image-processing features, enhancing functionality like documentation and claims processing.
- Prevented memory leaks and ANRs with Leak Canary and optimized app stability.
- Used Data Binding for dynamic interaction with the app's UI and optimized layouts using Fragments for better responsiveness.
- Implemented Android Keystore for secure cryptographic key storage.
- Utilized GitLab for continuous integration and maintained smooth testing across devices.
- Cached network data with OrmLite and displayed data through MPAndroidChart for improved performance.
- Incorporated a PDFViewer to enhance user experience for viewing documents.
- Performed Espresso tests and conducted code reviews to ensure high-quality, reusable code.

Jan 2013 – Dec 2014 **Software Engineer**
Oracle, Austin, TX

- Designed and implemented a healthcare management dashboard focused on usability, enhancing doctor and hospital information displays using React.js and modern JavaScript frameworks.
- Directed the seamless integration of Node.js backend services with MongoDB databases, optimizing data interactions across the application.
- Leveraged AWS cloud solutions, including EC2, Lambda, and DynamoDB, to deploy, scale, and maintain highly available services.
- Conducted robust testing strategies, including unit, integration, and end-to-end testing, reducing critical bugs by 20% and boosting code reliability by 30%.
- Played a key role in CI/CD implementation, streamlining automated testing and deployment for rapid and reliable updates.
- Collaborated with UI/UX teams and backend developers to integrate RESTful APIs, improving app functionality and user engagement.
- Advocated for Agile practices, reducing delivery times by 10%, and created comprehensive technical documentation that increased team efficiency by 25%.

Education:

Bachelor's, Computer Science

Master's, Computer Science

University of Central Missouri